L05: Interference

S&DS 6165: Topics in Causal Inference

Johan Ugander

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Lecture brief: View the source `brief.md` · How these notes are made

Lecture outline

- See why treating one person can change somebody else's outcome, and expand the potential-outcome table accordingly.
- Specify which comparisons matter and which assignments reveal them.
- Test for interference by preserving the comparisons that a null hypothesis makes observable.
- Design experiments to create informative exposures, balancing their frequency against dependence and possible bias.

1 A treatment changes an environment

Suppose a messaging platform changes its interface to encourage users to send more messages. A treated user sees the new interface. An untreated user may nevertheless receive more messages because their friends see it. The experiment changes both a person's own interface and the social environment in which that interface is used. Comparing treated and untreated users need not answer what would happen if the platform changed the interface for everyone.

The preceding lectures treated the potential outcomes of a unit as fixed once that unit's treatment was specified. We now retain the finite-population, design-based framework but enlarge the intervention being specified. The central difficulty is not that randomization stops working. It is that randomizing individual treatment statuses creates a distribution of social environments, and that distribution may differ substantially from the environment named by the scientific question.

1.1 What SUTVA lets us leave out

The **stable unit treatment value assumption**, or SUTVA, combines two requirements. First, a unit's outcome does not depend on other units' treatment assignments: there is no interference. Second, each treatment label describes an intervention sufficiently well that there are no outcome-relevant, hidden versions of that treatment. The first requirement is about which units' assignments matter; the second is about whether the assignments themselves adequately describe what is done.

For a finite population of N units, let $\mathbf{z} = (z_1, \ldots, z_N) \in \{0,1\}^N$ be a possible assignment vector. The potential outcome $Y_i(\mathbf{z})$ describes unit i under that entire intervention. All of these functions are fixed. The design draws the random vector $\mathbf{Z}$, and consistency gives

$$Y_i^{\text{obs}} = Y_i(\mathbf{Z}).$$

The no-interference restriction is

$$Y_i(\mathbf{z}) = Y_i(\mathbf{z}') \quad \text{whenever } z_i = z_i'. \tag{1}$$

Under this restriction, the assignments to the other $N-1$ units can be suppressed and we recover the two entries $Y_i(0)$ and $Y_i(1)$. Without it, a binary-treatment science table has N rows and 2^N

assignment columns. The table is large because the intervention is large, not because we have changed the definition of a potential outcome.

Consider two users. A hypothetical science table under SUTVA could be

Unit	Own control, $Y_i(0)$	Own treatment, $Y_i(1)$
User 1	2	3
User 2	4	7

Now allow messages from the other person to affect the outcome. For this example, specify the complete response functions as

$$Y_1(z_1, z_2) = 2 + z_1 + 2z_2, \qquad Y_2(z_1, z_2) = 4 + 3z_2 + z_1.$$

User 1 gains one unit from their own treatment and two from the other user's treatment; user 2 gains three from their own treatment and one from the other user's treatment. Transposing the expanded science table makes all four assignments easy to inspect:

Assignment (z_1, z_2)	$Y_1(z_1, z_2)$	$Y_2(z_1, z_2)$
$(0, 0)$	2	4
$(1, 0)$	3	5
$(0, 1)$	4	7
$(1, 1)$	5	8

If the realized assignment is $(1, 0)$, we observe the entire second row of this transposed table: outcomes 3 and 5. We do not observe one person's outcome from the all-treated world and the other's from the all-control world. This is the first change in how we should read experimental data under interference.

The other component of SUTVA matters too. "Use the new recommendation algorithm" is incomplete if its training data change with the experiment's allocation. One can expand the intervention to include the training protocol and thereby obtain well-defined potential outcomes, but that expansion does not restore no interference. A precise intervention and the absence of spillovers are separate requirements.

2 Four mechanisms that create interference

The mechanisms below produce different scientific questions. They also suggest different restrictions on $Y_i(\mathbf{z})$. The displayed functions are illustrative models developed for these notes, not claims that the cited applications obey an exact equation.

Social interaction. A reminder can affect its recipient and, through conversation, the recipient's friends. Bond et al. (2012) studied political mobilization through an online social network. The distinction between receiving a message and being connected to its recipients is central to this setting. Let A_{ij} indicate a pretreatment social connection from j to i. A simple model is

$$Y_i(\mathbf{z}) = a_i + b_i z_i + \sum_{j \neq i} A_{ij} c_{ij} z_j. \tag{2}$$

Here a_i is the baseline outcome, b_i the own-treatment contribution, and c_{ij} the contribution of treating a connected person. These coefficients are fixed, and the equation is a restriction on the science table. It is not a regression fitted to the observed cross-section. Social reinforcement could also generate interactions such as $z_j z_k$, so additivity is a substantive simplification.

Markets. Two sellers can compete for the same buyer even if they never communicate. A treatment that makes one listing more attractive can divert demand from another. Holtz et al. (2024) use a pricing meta-experiment on Airbnb to study this kind of interference. For a small illustration, let two sellers divide one unit of demand, and let $\delta > 0$ measure the increase in a seller's attractiveness under treatment:

$$Y_1(z_1, z_2) = \frac{\exp(\delta z_1)}{\exp(\delta z_1) + \exp(\delta z_2)}, \qquad Y_2(z_1, z_2) = 1 - Y_1(z_1, z_2).$$

A treated seller outperforms an untreated competitor. Yet changing both sellers' treatment leaves both market shares at one half. An individual advantage can therefore coexist with zero change in average market share under universal adoption. For revenues or welfare, total demand might also change; that would require a different response function. Calling a treatment successful requires specifying which market outcome and which policy comparison matter.

Spatial movement. A local restriction can shift activity across a border. Chang et al. (2023) study spillovers in mobility under California's county-level COVID-19 restrictions using a regression-discontinuity design. Their application is observational; it illustrates the mechanism, while our subsequent analysis concerns randomized assignments. A stylized county outcome could depend on both its own policy and neighboring policies:

$$Y_i(\mathbf{z}) = a_i - b_i z_i + \sum_{j \neq i} w_{ij} z_j, \qquad w_{ij} \geq 0.$$

Own restrictions reduce local activity, whereas restrictions elsewhere can redirect activity toward county i. The weights might reflect travel connections rather than geographic adjacency. Restricting one county and restricting an entire region are distinct interventions even for a resident whose own county has the same policy in both cases.

Shared learning and symbiosis bias. A recommendation algorithm learns from the behavior it induces. If two experimental algorithms share those data, one can benefit from exploration performed by the other. Brennan et al. (2025) study the resulting discrepancy between mixed-algorithm experiments and separate deployment. Let $D_t(\mathbf{z})$ denote the training data accumulated before period t under allocation $\mathbf{z}$, and let $\theta_{a,t}(\mathbf{z})$ be the fitted state of algorithm $a \in \{0, 1\}$. A schematic representation is

$$\theta_{a,t}(\mathbf{z}) = \mathcal{A}_a\{D_t(\mathbf{z})\}, \qquad Y_{it}(\mathbf{z}) = g_i\{z_i, \theta_{z_i,t}(\mathbf{z})\}.$$

The symbol $\mathcal{A}_a$ denotes the algorithm's training procedure. Although user i interacts with one algorithm, its trained state depends on other users' assignments. Separating training data can reduce

this channel, but also changes the quantity of data available to each algorithm. That intervention may itself differ from full deployment. The system being evaluated includes the learning process, not just a fixed scoring rule.

These examples are not exhausted by a friendship graph. A graph used in design should represent a plausible channel of causal influence: competition, mobility, shared learning, or social interaction. A convenient data structure alone is not evidence that all interference follows its edges.

3 Which treatment effect?

3.1 Universal adoption and individual treatment

For a decision about deployment to everyone, the natural finite-population target is

$$\tau_{\mathrm{G}} = \frac{1}{N}\sum_{i=1}^{N}\{Y_i(\mathbf{1}_N) - Y_i(\mathbf{0}_N)\}. \tag{3}$$

The vectors $\mathbf{1}_N$ and $\mathbf{0}_N$ assign everyone to treatment and control, respectively. We call τ_{G} the **global average treatment effect** (GATE); "total treatment effect" is another name used for this deployment contrast. It includes the consequences of treating the unit and of changing its environment. In the two-user table, it equals $\{(5-2)+(8-4)\}/2 = 3.5$.

Now independently treat each unit with probability $p \in (0,1)$. The ordinary individual-treatment HT contrast is

$$\hat{\tau}_{\mathrm{own}} = \frac{1}{N}\sum_{i}\left\{\frac{Z_i Y_i^{\mathrm{obs}}}{p} - \frac{(1-Z_i)Y_i^{\mathrm{obs}}}{1-p}\right\}.$$

Its expectation can still be calculated exactly. Conditioning first on the other units' assignments, independence gives

$$\mathbb{E}_Z(\hat{\tau}_{\mathrm{own}}) = \frac{1}{N}\sum_{i}\mathbb{E}_{Z_{-i}}\{Y_i(1,\mathbf{Z}_{-i}) - Y_i(0,\mathbf{Z}_{-i})\} \equiv \tau_D(p). \tag{4}$$

Here $\mathbf{Z}_{-i}$ is the vector with unit i removed, and $Y_i(a,\mathbf{Z}_{-i})$ places a in position i. The expectation averages over independent Bernoulli-p assignments to the others. Thus $\tau_D(p)$ changes one's own treatment while averaging over the same distribution of other people's treatments on both sides. It is a design-specific average direct effect, not generally the universal-adoption contrast.

In the two-user example, $\tau_D(p) = (1+3)/2 = 2$ for every p. The difference from 3.5 is the average spillover contribution, 1.5. In other response functions, own-treatment effects interact with others' treatments and $\tau_D(p)$ itself changes with p. Neither randomization nor inverse-probability weighting says these two estimands must coincide.

This calculation uses independent assignment. Under complete randomization, conditioning on a unit being treated leaves fewer treatments available for its peers than conditioning on it being a control. One must use the actual conditional assignment laws. The design is part of the interpretation of an individual-treatment contrast under interference.

3.2 Exposure mappings compress the science table

A messaging feature may depend on the assignments of a person's close contacts but not on assignments elsewhere. Or only the number of treated contacts may matter, rather than their identities. These are different restrictions, and **exposure mappings** give a language for stating them (Aronow and Samii 2017).

For each unit, an exposure mapping is a specified function

$$f_i : \{0,1\}^N \longrightarrow \mathcal{E}_i, \qquad E_i = f_i(\mathbf{Z}),$$

where $\mathcal{E}_i$ is the set of possible exposure labels. The substantive assumption is that assignments receiving the same label produce the same outcome:

$$f_i(\mathbf{z}) = f_i(\mathbf{z}') \quad \Longrightarrow \quad Y_i(\mathbf{z}) = Y_i(\mathbf{z}'). \tag{5}$$

We can then write the common outcome as $y_i(e)$ for exposure e. Lowercase $y_i(e)$ distinguishes an exposure-indexed outcome from $Y_i(\mathbf{z})$, the original assignment-indexed function. The mapping is an assumption about equality among cells of the science table. Computing it from observed assignments does not establish that the equality holds.

Let G be a fixed, pretreatment, undirected interference graph. Write $\mathcal{N}_i$ for the **closed neighborhood** of i: the unit itself and its neighbors. Its size is $d_i = |\mathcal{N}_i|$. A local-interference model says that $Y_i(\mathbf{z})$ depends only on the subvector $\mathbf{z}_{\mathcal{N}_i}$. This permits arbitrary interactions among neighbors' assignments. A more restrictive mapping uses own treatment and the fraction of other neighbors treated:

$$f_i(\mathbf{z}) = (z_i, q_i(\mathbf{z})), \qquad q_i(\mathbf{z}) = \frac{1}{d_i - 1} \sum_{j \in \mathcal{N}_i \setminus \{i\}} z_j,$$

with $q_i = 0$ for an isolated unit. Equal fractions now imply equal outcomes even when different neighbors are treated. A close friend and a distant acquaintance become interchangeable for this purpose. That is additional modeling, not a consequence of locality.

For GATE estimation, two exposure conditions have a special role. Unit i is **network exposed to treatment** if its observed outcome equals $Y_i(\mathbf{1}_N)$, and **network exposed to control** if it equals $Y_i(\mathbf{0}_N)$. Under the local-interference model, sufficient observable conditions are

$$E_i = e_1 : \ z_j = 1 \text{ for every } j \in \mathcal{N}_i, \qquad E_i = e_0 : \ z_j = 0 \text{ for every } j \in \mathcal{N}_i. \tag{6}$$

These are full-neighborhood exposures (Ugander et al. 2013). They allow a local part of a mixed experiment to reveal an outcome from a global counterfactual. The outcome may vary arbitrarily among mixed neighborhoods; we need not pool all mixed assignments into one outcome-equivalent category. Threshold or fractional exposure rules can make the extreme labels more common, but identifying them with global outcomes then requires stronger substantive assumptions.

There are other ways to restrict the science table. An exposure model equates outcomes under different assignments. A low-order response model instead restricts how treatment indicators combine

algebraically, potentially allowing every assignment to have a distinct outcome. L06 develops that alternative. Neither approach obtains its assumptions from randomization alone.

3.3 Moving SUTVA to clusters

Sometimes a plausible grouping is available before specifying a network: households, schools, or geographically isolated villages. Let $\mathcal{C} = \{C_1, \dots, C_K\}$ be a partition and $c(i) \in \{1, \dots, K\}$ the index of the cluster containing unit i. **Partial interference** assumes that

$$\mathbf{z}_{C_{c(i)}} = \mathbf{z}'_{C_{c(i)}} \quad \Longrightarrow \quad Y_i(\mathbf{z}) = Y_i(\mathbf{z}'). \tag{7}$$

Assignments within the cluster can interact arbitrarily; assignments outside it do not affect i. Hudgens and Halloran (2008) develop effects and designs in this setting. Karrer et al.'s network-experimentation paper, first circulated in 2020, calls this SUTVA for clusters (Karrer et al. 2021). It is a restriction on the interference structure, not an automatic consequence of assigning treatments by cluster. Candogan, Chen, and Niazadeh (2024) also use no interference across clusters when optimizing correlations among cluster assignments. This connects cluster SUTVA to an established literature on partial interference and to subsequent work on experimental design.

If each cluster receives a single treatment W_k, the allowed unit assignments satisfy $Z_i = W_{c(i)}$. On this restricted set of assignments, partial interference implies that the outcome depends only on $W_{c(i)}$. Cluster totals can then be analyzed as potential outcomes indexed by one cluster treatment, with size weighting if the target averages over individuals.

There is a useful distinction between requiring partial interference for every conceivable assignment and requiring this invariance only on the design's cluster-constant assignments. For example, suppose units 3 and 4 form another cluster and $Y_1(\mathbf{z}) = z_1 + z_3 - z_4$. Cross-cluster interference exists when units 3 and 4 receive different treatments. Under a design that always assigns them together, their effects cancel. The restricted experiment therefore has a cluster-level invariance that the unrestricted science table lacks. This example illustrates a distinction between assumption domains; it is not a new definition attributed to Karrer et al.

Cluster SUTVA can be operationally useful without committing to a detailed exposure mapping within each cluster. Mathematically, however, it can itself be expressed as a mapping that retains the cluster's treatment vector. Exposure mappings and cluster SUTVA are therefore overlapping ways to express restrictions, not mutually exclusive causal frameworks.

4 From observed exposures to an estimator

4.1 The inverse probability is now an exposure probability

Suppose the mapping is adequate for exposures e_1 and e_0, so that $y_i(e_a) = Y_i(a\mathbf{1}_N)$ for $a \in \{0, 1\}$. Define the marginal and joint exposure probabilities under the chosen design by

$$\pi_i(e) = \mathbb{P}_Z(E_i = e), \qquad \pi_{ij}(e, e') = \mathbb{P}_Z(E_i = e, E_j = e').$$

These probabilities average over assignments, holding the graph and potential outcomes fixed. They are not estimated observational propensity scores. A unit can have treatment probability one half and an exceedingly small probability of experiencing an all-treated neighborhood.

If both required exposures have positive probability for every unit, define

$$\hat{\mu}_{\mathrm{HT}}(e) = \frac{1}{N}\sum_i \frac{\mathbf{1}\{E_i = e\}Y_i^{\mathrm{obs}}}{\pi_i(e)}, \qquad \hat{\tau}_{\mathrm{G,HT}} = \hat{\mu}_{\mathrm{HT}}(e_1) - \hat{\mu}_{\mathrm{HT}}(e_0). \tag{8}$$

Proposition 4.1 (Design unbiasedness under an adequate exposure model)**.** *If exposure e_a reveals $Y_i(a\mathbf{1}_N)$ and $\pi_i(e_a) > 0$ for every i and $a \in \{0,1\}$, then $\mathbb{E}_Z(\hat{\tau}_{\mathrm{G,HT}}) = \tau_{\mathrm{G}}$.*

Proof. On the event $E_i = e_a$, consistency and the exposure assumption replace the observed outcome by the fixed number $Y_i(a\mathbf{1}_N)$. Consequently,

$$\mathbb{E}_Z\left\{\frac{\mathbf{1}\{E_i = e_a\}Y_i^{\mathrm{obs}}}{\pi_i(e_a)}\right\} = \frac{\pi_i(e_a)Y_i(a\mathbf{1}_N)}{\pi_i(e_a)} = Y_i(a\mathbf{1}_N).$$

Average over units and subtract the control expression. No independence among exposures is used. □

The familiar HT cancellation survives interference. What changes is the event that makes a particular potential outcome observable. Weighting cannot repair a misspecified exposure model: if different assignments with label e_1 produce different outcomes, the weighted average targets their design-weighted mean, which need not equal the all-treated outcome.

A Hájek version divides the weighted outcome sum by the weighted count of units in that exposure. As in L02, this can improve stability, but exact design unbiasedness is generally lost and a zero denominator must be handled. Choosing normalization does not resolve an incorrect mapping or zero exposure probability.

4.2 Positivity and joint observability

Under independent Bernoulli-p unit assignment and full-neighborhood exposure,

$$\pi_i(e_1) = p^{d_i}, \qquad \pi_i(e_0) = (1-p)^{d_i}. \tag{9}$$

Each of the d_i assignments must agree. At $p = 1/2$, a closed neighborhood of size 11 receives either particular extreme exposure with probability $1/2048$. A large experiment can contain many treated units but very few observations of the outcomes needed for its deployment target. This is L04's weak-positivity problem created by the combination of network structure, estimand, and design.

The pairwise probabilities matter for inference. Write $s_{e_1} = 1$ and $s_{e_0} = -1$. Expanding the variance of Equation 8 gives

$$\mathrm{Var}_Z(\hat{\tau}_{\mathrm{G,HT}}) = \frac{1}{N^2}\sum_{i,j}\sum_{e,e' \in \{e_1,e_0\}} s_e s_{e'} \frac{\pi_{ij}(e,e') - \pi_i(e)\pi_j(e')}{\pi_i(e)\pi_j(e')} y_i(e) y_j(e'). \tag{10}$$

For $i = j$, interpret the joint probability in the usual way: it equals $\pi_i(e)$ if $e = e'$ and zero otherwise. The formula follows by writing the estimator as a sum of weighted indicators and using the covariance of each pair. It displays both rare-exposure weights and dependence among exposures.

Even when each marginal probability is positive, some joint probabilities are zero. If two closed neighborhoods overlap, one cannot simultaneously be all treated and the other all control. The corresponding product of potential outcomes is never jointly observed. It cannot be estimated by dividing an observed product by its joint probability. Conservative variance procedures handle these unidentified terms with bounds; normal approximations additionally require an appropriate sequence of growing experiments and restrictions on dependence (Aronow and Samii 2017). An ordinary independent-observation standard error ignores precisely the structure that made interference relevant.

5 Testing whether interference is present

An assumption can be useful without being beyond scrutiny. Testing interference requires care because "no interference" still permits each unit to have an unknown direct treatment effect. We must preserve that uncertainty while finding assignments under which the relevant outcomes can be imputed.

5.1 Make the null sharp on a conditional experiment

The global sharp null $Y_i(\mathbf{z}) = Y_i(\mathbf{z}')$ for every $i, \mathbf{z}, \mathbf{z}'$ rules out all treatment effects. Rejecting it does not specifically establish interference: ordinary direct effects suffice. The no-interference null in Equation 1 is weaker. It supplies no value for a treated person's outcome under their own control assignment.

The conditional-testing approach of Athey, Eckles, and Imbens (2018) makes the null sharp for selected **focal units** by restricting the assignments considered. Choose a set F of focal units before seeing treatment or outcomes, for example using only the pretreatment graph. Condition on their realized own assignments $\mathbf{Z}_F$. Under no interference, their outcomes are then fixed for every reassignment of the remaining, auxiliary units.

To see a statistic one could use, let $q_i(\mathbf{z})$ be the treated-neighbor fraction defined earlier, and let $\overline{Y}_F^{\text{obs}}$ be the mean focal outcome. Consider

$$T(\mathbf{z}) = \sum_{i \in F} (Y_i^{\text{obs}} - \overline{Y}_F^{\text{obs}}) \left[q_i(\mathbf{z}) - \mathbb{E}_Z\{q_i(\mathbf{Z}) \mid \mathbf{Z}_F\} \right].$$

The outcome factors stay fixed. The exposure factors vary as auxiliary treatments are reassigned. Positive values indicate that focal outcomes tend to be larger where randomized neighbor exposure is larger. A two-sided test could use $|T|$.

The reference distribution must be the actual conditional design, $\mathbb{P}_Z(\mathbf{Z}_{-F} \mid \mathbf{Z}_F)$. Under independent Bernoulli assignment, auxiliary assignments remain independent with probability p. Under complete randomization, the number of auxiliary treatments is fixed by the original treatment budget and the

observed focal assignments. Simply flipping independent coins would analyze the wrong experiment in the latter case.

Conditional on $\mathbf{Z}_F$, the observed assignment and a fresh draw from this conditional law have the same distribution under the null, while every outcome entering the statistic is known. Ranking the observed statistic in this distribution therefore gives a valid randomization p-value, with discreteness handled conservatively. For Monte Carlo sampling, including the observed allocation in the usual plus-one construction avoids zero estimated p-values.

This argument does not require the chosen neighbor statistic to describe the actual interference correctly. Under the null there is no interference to describe. A poor graph or statistic reduces power, whereas an incorrect conditional assignment law can invalidate the test. Choosing focal units after observing assignments is possible, but requires conditioning on the selection mechanism as well; selecting favorable focal units and then pretending they were fixed is not justified.

5.2 Randomize the randomization mechanism

A different strategy deliberately creates different local environments at the same marginal treatment probability. Saveski et al. (2017) compare individual and cluster randomization within a randomized meta-experiment. The outer randomization allocates groups to the two experimental strategies; the inner randomization assigns treatment according to each strategy. Under no interference, both strategies target the same average direct effect. Their discrepancy can reveal interference without requiring an exact spillover model.

For a concrete design-based version, let $R_i \in \{0, 1\}$ record the strategy to which unit i is assigned, with known probability $\rho_i(r) = \mathbb{P}(R_i = r)$. Suppose each strategy gives that unit conditional treatment probability p. An HT contrast for strategy r is

$$\hat{\tau}_r = \frac{1}{N} \sum_i \frac{\mathbf{1}\{R_i = r\}}{\rho_i(r)} \left\{ \frac{Z_i Y_i^{\text{obs}}}{p} - \frac{(1 - Z_i) Y_i^{\text{obs}}}{1 - p} \right\}.$$

Under no interference, iterated expectation over the outer and inner assignments gives $\mathbb{E}(\hat{\tau}_r) = N^{-1} \sum_i \{Y_i(1) - Y_i(0)\}$ for both strategies. Hence $\hat{\tau}_1 - \hat{\tau}_0$ is centered at zero. This calculation explains the role of randomizing strategy membership: the comparison is not confounded by choosing different populations for the two designs. Its variance must incorporate both randomizations and their covariances.

Under interference, cluster assignment makes own treatment more correlated with neighbors' treatments. In the two-user example, assigning both users together produces the global contrast 3.5, whereas independent assignment produces the direct-effect contrast 2. A strategy comparison can detect this discrepancy. For a real network with edges crossing clusters, neither strategy need identify GATE. The test compares the effects revealed by the two designs; it does not certify one as unbiased for universal deployment. Nonrejection likewise does not establish no interference, because the designs can miss relevant channels or have little power.

5.3 Use randomized increases in treatment allocation

Platforms often expand a treatment gradually. A control unit that remains untreated can experience an increasingly treated environment. Han, Li, Mao, and Wu (2023) use increasing-allocation experiments to test for cross-unit interference. The useful variation is in other units' randomized treatment histories, rather than a raw comparison of outcomes at different dates.

Consider two periods with marginal treatment probabilities $p_1 < p_2$. Draw independent $U_i \sim \text{Uniform}(0,1)$ and set $Z_{it} = \mathbf{1}\{U_i \leq p_t\}$. Each unit's treatment history is one of $(0,0)$, $(0,1)$, or $(1,1)$, with probabilities $1-p_2$, p_2-p_1, and p_1. The history $(1,0)$ is impossible. Write $Y_{it}(\mathbf{z}_{1:2})$ for the potential outcome when the full two-period assignment matrix is specified. The no-cross-unit-interference null says that this outcome depends only on unit i's own treatment history. It can still depend on time and on that own history.

A conditional experiment can hold focal units' entire treatment histories fixed and randomize the auxiliary histories. For example, preselect a focal set F using the graph and condition on the histories of **every unit in this original set**. Within the statistic, use only focal units whose own assignment is unchanged, and form $\Delta Y_i = Y_{i2}^{\text{obs}} - Y_{i1}^{\text{obs}}$. These differences are fixed under the null after conditioning on focal histories. A statistic can associate ΔY_i with the change in treated-neighbor fraction, $q_i(\mathbf{Z}_2) - q_i(\mathbf{Z}_1)$. Focal units that switch treatment remain conditioned upon even though their outcomes do not enter this statistic; they do not move into the auxiliary pool.

Conditional on the counts of histories among units outside the original F, iid rollout assignment makes those auxiliary histories exchangeable. Permuting **whole histories only among units outside** F preserves both the rollout and this conditional law. Recompute neighbor exposures after each permutation and compare the observed statistic with the resulting distribution. Separately permuting the two treatment columns could create forbidden $(1,0)$ histories and would not reproduce the design.

Differencing can help remove persistent unit differences from the statistic; its validity here does not require all time trends to be common because focal outcomes are held fixed under the null. Tests that instead exchange periods, or interpret a changing treatment-control contrast as evidence by itself, require additional restrictions. The paper also develops a more powerful procedure under a time-fixed-effect model. That additional model is consequential: seasonality, changing treatment effects, or carryover can otherwise resemble an allocation effect.

The three strategies use distinct sources of variation. Conditional tests hold own treatment fixed while varying others' assignments. Meta-experiments vary the mechanism assigning treatment. Rollouts vary treatment histories across time. In each case, valid inference follows from an explicit assignment or invariance argument; the visual appearance of spillovers is only a starting point.

6 Designing for informative exposures

Testing can reveal that individual randomization answers an incomplete policy question. The next design question is how to make the desired exposures occur more often while retaining enough independent comparisons.

6.1 Graph cluster randomization

For a fixed partition $\mathcal{C}$, assign cluster treatments $W_1, \ldots, W_K$ independently as Bernoulli-p, and set $Z_i = W_{c(i)}$. Let $k_i(\mathcal{C})$ be the number of distinct clusters intersecting the closed neighborhood $\mathcal{N}_i$. Then full-neighborhood exposure probabilities are

$$\pi_i(e_1 \mid \mathcal{C}) = p^{k_i(\mathcal{C})}, \qquad \pi_i(e_0 \mid \mathcal{C}) = (1-p)^{k_i(\mathcal{C})}. \tag{11}$$

Only the assignments of these clusters must agree. The number of independent choices in the exponent can be much smaller than the number of units. This is the central mechanism of graph cluster randomization (GCR) (Ugander et al. 2013). It does not require there to be no edges across clusters: an exposure estimator can account for those edges when calculating the events and their probabilities.

Take a path of six units, $1-2-3-4-5-6$, and $p = 1/2$. Under individual randomization, each endpoint has closed-neighborhood size two and each interior unit size three. Group the path into clusters $\{1,2,3\}$ and $\{4,5,6\}$.

Unit	Closed-neighborhood size d_i	Individual $\pi_i(e_1)$	Clusters touched k_i	Cluster $\pi_i(e_1)$
1 and 6	2	1/4	1	1/2
2 and 5	3	1/8	1	1/2
3 and 4	3	1/8	2	1/4

Interior units away from the cut benefit most. Units 3 and 4 remain near a cluster boundary and require both clusters to agree. The calculation makes clear why counting the number of cut edges alone need not describe every unit's statistical difficulty.

Larger clusters can increase these marginal probabilities but also increase dependence. With one cluster containing the entire graph, every unit receives an extreme exposure, yet there is only one randomized treatment coin. The observed data are either the whole all-treated population or the whole all-control population. HT remains unbiased across repeated experiments, but the two worlds are never observed together. Exposure frequency and precision are different objectives.

For unequal clusters, one must also distinguish averaging people from averaging clusters. The unweighted mean of cluster means targets an average cluster, giving a person in a small cluster more weight. An estimator of Equation 3 must preserve the intended individual weights.

6.2 Randomize the clustering too

In a fixed clustering, some units repeatedly occupy favorable interiors and others unfavorable boundaries. Randomized graph cluster randomization (RGCR) draws a partition from a specified distribution Q, then randomizes cluster treatments. Ugander and Yin (2023) study this design and randomized graph-decomposition algorithms, with variance guarantees under graph-structure assumptions.

The two-stage exposure probability is

$$\pi_i(e_1) = \mathbb{E}_{\mathcal{C}\sim Q}\{p^{k_i(\mathcal{C})}\}, \qquad \pi_i(e_0) = \mathbb{E}_{\mathcal{C}\sim Q}\{(1-p)^{k_i(\mathcal{C})}\}. \tag{12}$$

The expectation is over the random partition. For example, if a unit's neighborhood touches one cluster half the time and three clusters half the time, its treatment-exposure probability at $p = 1/2$ is $(1/2)(1/2) + (1/2)(1/8) = 5/16$. It is not permanently assigned the unfavorable probability $1/8$.

One simple partition construction gives units independent continuous random priorities and assigns each unit to the highest-priority vertex in its closed neighborhood; equal chosen vertices define a cluster. This is the 1-hop-max construction. Its purpose is to make cluster membership vary through local random choices while remaining tied to graph proximity. The resulting clusters are then independently treated or controlled.

An exposure estimator using the marginal probabilities in Equation 12 is unbiased over **both** randomizations. It need not be unbiased conditional on the realized partition. Using conditional probabilities instead defines another estimator, with conditional unbiasedness when those probabilities are positive. The analyst must match the probability in the denominator to the randomization distribution used for the claimed result.

Monte Carlo computation can approximate the marginal probabilities by repeatedly drawing partitions. This is design simulation and does not require outcomes. However, replacing a probability by a noisy estimate inside its reciprocal does not preserve exact unbiasedness. Rare exposures require accurate probability calculations and enough simulations; increasing the number of outcome units does not automatically eliminate probability-computation error.

7 A clustering is optimal for an objective

Community detection seeks a descriptive partition of a network. Experimental design asks which partition makes an estimator accurate for a specified causal target. These problems can overlap, but their objectives need not agree.

Viviano et al. (2026) study clustering for the particular assignment-weighted contrast

$$\hat{\tau}_{\mathcal{C}} = \frac{2}{N}\sum_i (2Z_i - 1)Y_i^{\text{obs}}, \tag{13}$$

under independent fair cluster assignment. The fixed denominator is N, not a realized treatment-arm count. This estimator weights own treatment, rather than requiring full-neighborhood exposure, and can be biased for GATE when clusters exchange spillovers. Their framework chooses clusters by a worst-case bias-variance criterion over a restricted class of potential outcomes, leading to a penalized graph-cut optimization. The result is therefore about an estimator, target, design class, and response class together. It is not a claim that a single partition is optimal for all analyses.

An additive calculation shows why this criterion has a different form from maximizing exposure probabilities. Use Equation 2, and abbreviate $\gamma_{ij} = A_{ij}c_{ij}$. Then

$$\tau_{\mathrm{G}} = \frac{1}{N}\sum_i \left(b_i + \sum_{j \neq i} \gamma_{ij} \right).$$

Under fair cluster assignment, $\mathbb{E}_Z(2Z_i - 1) = 0$. Also,

$$2\mathbb{E}_Z\{(2Z_i - 1)Z_j\} = \begin{cases} 1, & c(j) = c(i), \\ 0, & c(j) \neq c(i). \end{cases}$$

For a same-cluster unit, $Z_j = Z_i$; for a different-cluster unit, the two fair assignments are independent. Substituting into the estimator gives

$$\mathbb{E}_Z(\hat{\tau}_{\mathcal{C}}) - \tau_{\mathrm{G}} = -\frac{1}{N}\sum_i \sum_{j : c(j) \neq c(i)} \gamma_{ij}. \tag{14}$$

The estimator captures within-cluster contributions because they move with own treatment. Across-cluster contributions cancel from its expected treatment-control comparison. For nonnegative spillovers it underestimates GATE by the omitted contribution. With mixed signs, cancellation can make bias smaller or reverse its direction.

Variance pushes the partition in the opposite direction. Even in the simple case $Y_i(\mathbf{z}) = a_i$ with no effects, the estimator is random. Writing $A_k = \sum_{i \in C_k} a_i$,

$$\mathrm{Var}_Z(\hat{\tau}_{\mathcal{C}}) = \frac{4}{N^2}\sum_{k=1}^{K} A_k^2.$$

If all baseline outcomes equal a constant a, this is $4a^2N^{-2}\sum_k |C_k|^2$. Combining two clusters of sizes u and v increases the sum of squared sizes by $2uv$, even if it eliminates a cut. A design criterion must weigh the expected importance of spillovers against this loss of independent variation. Covariate adjustment can change the variance calculation by replacing predictable baseline components with residuals.

The preceding calculations are a pedagogical special case. In the causal-clustering analysis, bounds on spillover strength and outcome magnitudes, along with local asymptotic restrictions, justify a corresponding worst-case optimization. Tuning the relative bias and variance penalty encodes how large relevant spillovers could be. Pilot evidence or subject knowledge can inform it, and sensitivity across plausible penalties is more informative than treating the choice as a purely graph-theoretic constant.

Changing to the exposure HT estimator changes the problem. Under a correct mapping, its bias is zero and the design concern becomes its joint exposure probabilities and variance. A partition that is attractive for Equation 13 need not be attractive for Equation 8. This is a concrete instance of designing an experiment and its analysis together.

8 Advanced material: designs built around incompatible exposures

Clustering begins by asking which units should share an assignment. Another starting point is to ask which outcomes needed for the estimand cannot be observed together. This makes the estimand itself part of the design's combinatorial structure.

Fix two desired neighborhood configurations for each unit, indexed by $a = 0, 1$, and let their potential outcomes be $y_i(e_{ia})$. The target is $N^{-1} \sum_i \{y_i(e_{i1}) - y_i(e_{i0})\}$. For GATE these configurations are all treated and all control. For a direct effect at untreated neighbors, they differ only in the focal unit's own assignment.

Two units conflict if **some** pair of their desired configurations cannot be realized by a single assignment vector. The conflict graph places an edge between such units, including a self-loop because one unit cannot receive both distinct configurations simultaneously. Kandiros, Pipis, Daskalakis, and Harshaw (2026) use this graph to construct a design matched to the contrast.

On the path $1 - 2 - 3 - 4$, units 1 and 3 are not adjacent. Their closed neighborhoods nevertheless overlap at unit 2. Giving unit 1 its all-treated exposure requires $Z_2 = 1$, while giving unit 3 its all-control exposure requires $Z_2 = 0$. They conflict for GATE. In general, the GATE conflict graph connects vertices at graph distance at most two. For the direct-effect contrast that holds all neighbors untreated, units 1 and 3 can receive either desired configuration together: both require unit 2 to be untreated. For this contrast, only neighbors in the original graph conflict. The same physical network creates different observability constraints for different estimands.

The conflict-graph design first generates sparse requests for desired exposures, then resolves conflicting requests using a suitable priority ordering. Accepted requests force the necessary local treatment assignments. Its estimator uses the event that a request is accepted, not every accidental occurrence of the same exposure. If B_{ia} is that recorded event and $q_{ia} = \mathbb{P}_Z(B_{ia}) > 0$, the form is

$$\hat{\tau}_{\mathrm{CG}} = \frac{1}{N} \sum_i \left\{ \frac{\mathbf{1}\{B_{i1}\} Y_i^{\mathrm{obs}}}{q_{i1}} - \frac{\mathbf{1}\{B_{i0}\} Y_i^{\mathrm{obs}}}{q_{i0}} \right\}.$$

Accepted request B_{ia} guarantees the desired outcome, so the same HT cancellation proves unbiasedness. Restricting the counted events makes their dependence more tractable. Under bounded average squared target outcomes, the paper obtains a variance bound of order $\lambda(H)/N$, where $\lambda(H)$ is the largest eigenvalue of the conflict graph's adjacency matrix, including its self-loops. This guarantee belongs to the proposed design and modified estimator together; arbitrary prioritization or the ordinary exposure HT estimator does not inherit it automatically.

For intuition about the rate, a bounded-size set of conflicts around each unit allows $\lambda(H)$ to remain bounded and hence permits a $1/N$ variance bound. A complete conflict graph has $\lambda(H) = N$, so this bound no longer tends to zero. Merely increasing the number of outcomes does not create independent counterfactual comparisons when all desired exposures compete with one another.

9 Returning to the deployment question

Interference changes the relationship between what is assigned, what is observed, and what is wanted. The two-user science table already contained the essential issue: a person observed in a mixed assignment need not reveal their outcome under universal treatment or universal control.

An exposure assumption states when a local assignment reveals a global counterfactual. Exposure probabilities then turn this statement into an estimator, and joint probabilities govern its uncertainty. Tests can probe aspects of the assumption by designing conditional comparisons with imputable outcomes. Finally, experimental designs can make informative exposures more common, reduce the bias of a simpler estimator, or coordinate otherwise incompatible comparisons. Those are distinct statistical objectives, even when all three lead us to use a graph.

For a new application, the order of reasoning is therefore consequential: describe the mechanism and the deployment decision; state the target contrast; decide which restrictions on the science table are defensible; and choose an estimator and design that exploit those restrictions. L06 continues with restrictions on the form of the response function, where mixed assignments may be informative even without exact exposure equivalence.

10 Further reading

Hudgens and Halloran (2008) develop direct, indirect, total, and overall effects under partial interference, emphasizing that an allocation policy is itself an intervention. Aronow and Samii (2017) provide the design-exposure-estimand framework and the associated issues in variance estimation.

Athey, Eckles, and Imbens (2018), Saveski et al. (2017), and Han et al. (2023) give complementary routes to testing. Reading them together clarifies the distinction between sharpening a null by conditioning and changing the experiment to make its implications easier to detect.

Ugander et al. (2013) and Ugander and Yin (2023) develop fixed and randomized graph cluster designs. Eckles, Karrer, and Ugander (2017) analyze design and estimation as tools for reducing interference bias under response structures that need not satisfy a simple local exposure model. Karrer et al. (2021) discuss the practical interplay of clustering, regression adjustment, and experimentation at scale.

Viviano et al. (2026) and Kandiros et al. (2026) illustrate two different ways to turn a causal objective into a design problem: optimize a bias-variance criterion for a specified estimator, or coordinate estimand-specific observability constraints. Their guarantees should be compared with their estimator and response assumptions kept in view.

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